ADRIÀ MARTÍNEZ GALLIFA

Junior Backend Software Engineer

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PROJECTS

missme, Memory Hierarchy Simulator (Final Degree Project) *GitHub*

Rust, Vue, Tauri, TypeScript, RISC-V *2026*

- Built a desktop simulator of memory hierarchies (multiple cache levels, Von Neumann or Harvard, TLB, multilevel page tables, frames and swap) that runs user-written RISC-V programs.
- Kept the Rust engine and the Vue interface strictly separate, connected only through Tauri commands and events, so the engine could be tested standalone.
- Modelled each memory access as an explicit state machine, so execution could advance by atomic action, by instruction or continuously while every structure stayed inspectable.
- Modelled latencies so the measured AMAT could be compared against the one calculated by hand.
- Validated with 771 automated tests (469 for the Rust engine, 302 Vitest for the interface; 91% engine line coverage) and 7 exercises checked by hand against the theory.

Multilayer Epidemic Replication System *GitHub (team of 2)*

Python, TCP, WebSockets, Vue 3 *2025 - 2026*

- Implemented a replicated key-value store with eventual consistency across 7 nodes in 3 layers (core, L1, L2): main-backup replication, with upward flushes by write count (core to L1, after 10 writes) and by time (L1 to L2, every 10 s).
- Implemented Ricart-Agrawala distributed mutual exclusion for writes in the core layer, plus a FIFO reader-writer lock.
- Designed a custom binary TCP protocol and a real-time Vue 3 dashboard over WebSockets that shows the state of every node.

Compiler for a Custom Language *GitHub (team of 5)*

Java, MIPS *2025*

- Built a compiler for a custom domain-specific language, covering lexical, syntactic and semantic analysis and code generation, producing executables for MIPS.

EXPERIENCE

La Salle - Universitat Ramon Llull *Barcelona, Spain*

Teaching Assistant *2023 - 2026*

- Assisted Advanced Programming and Data Structures (PAED, Java, 2023–2026), Operating Systems (C, 2025–2026) and Advanced Operating Systems (2024–2025), about 120 students per course each year; the first year was a departmental collaboration.
- Chosen by the Operating Systems professor for the position through a selective application process.
- Designed and implemented 8 of the 30 lab exercises for the 2026 Operating Systems course, covering signals, threads, fork, sockets, select, mutexes, semaphores and shared memory with pipes.
- Answered technical questions in and outside the lab, graded assignments and projects, prepared teaching material and delivered about 5 full-hour lectures.

Software research line, La Salle *Barcelona, Spain*

Research Collaborator *2024 - 2026*

- Took part in a systematic literature review on green code (2014–2024), screening over 2,000 papers and analysing about 40 of the 120 read in full; attended the group's seminars.

EDUCATION

BSc in Computer Engineering *Barcelona, Spain*

La Salle - Universitat Ramon Llull *2021 - 2026*

ABOUT ME

Software engineering graduate (2026) with a backend focus and a strong interest in system design and architecture. Experience in systems programming, distributed systems, compilers and teaching.

SKILLS

- Languages: Rust, Java, C, C#, C++, Python, TypeScript, JavaScript, PHP, Bash
- Frontend & frameworks: Vue, React, Tauri, HTML/CSS
- Databases: PostgreSQL, SQLite
- Tools & DevOps: Docker, Git, npm, CMake, AWS (basic), AI coding assistants
- Systems: Linux (Ubuntu, Arch), Windows
- Editors: Neovim, VS Code, IntelliJ IDEA
- Certifications: Cisco Networking Academy, CCNA - Introduction to Networks (first module)

LANGUAGES

Catalan • Native
Spanish • Native
English • B2